

Master Math for JEE Main & JEE Advanced

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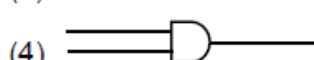
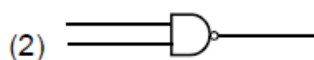
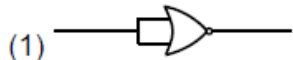
JEE Mains 2020 Jan Chapter wise Question Bank

Experimental Physics

Q1

Which of the following gate is reversible

इनमें से कौनसा द्वार उत्क्रमणीय है।

7th Jan Morning

Sol

(1)

A logic gate is reversible if we can recover input data from the output

eg. NOT gate

Q2

A battery of unknown emf connected to a potentiometer has balancing length 560 cm. If a resistor of resistance 10Ω is connected in parallel with the cell the balancing length change by 60 cm. If the

internal resistance of the cell is $\frac{n}{10}\Omega$, the value of 'n' is

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12

Let the emf of cell is $\mathcal{E}$ internal resistance is ' r ' and potential gradient is x .

माना कि सेल का विद्युत वाहक बल $\mathcal{E}$, आन्तरिक प्रतिरोध ' r ' तथा विभव प्रवणता x है।

only cell connected :

केवल बैटरी जुड़ी हो

$$\mathcal{E} = 560 x \quad \dots\dots(1)$$

After connecting the resistor

प्रतिरोध जोड़ने के पश्चात्

$$\frac{\mathcal{E} \times 10}{10 + r} = 500x \quad \dots\dots(2)$$

from (1) and (2)

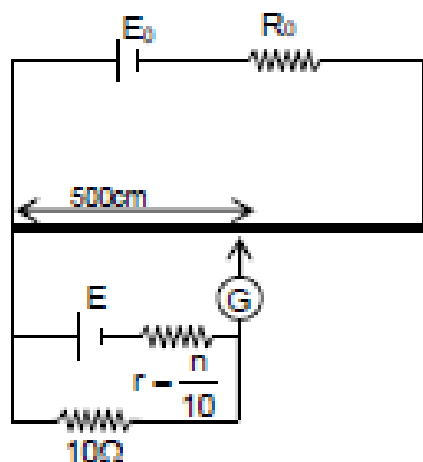
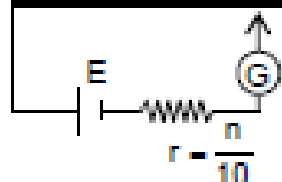
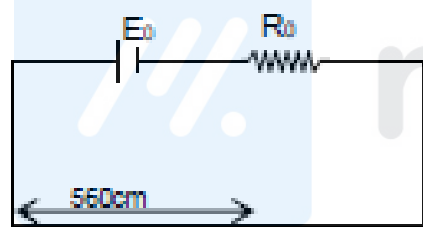
(1) तथा (2) से

$$\frac{560 \times 10}{10 + r} = 500 \mathcal{E}$$

$$56 = 50 + 5r$$

$$r = \frac{6}{5} = 1.2 \Omega$$

$$n = 12$$



Q3

There is a potentiometer wire of length 1200 cm and 60 mA current is flowing in it. A battery of emf 5V and internal resistance of 20Ω is balanced on potentiometer wire with balancing length 1000 cm. The resistance of potentiometer wire is—

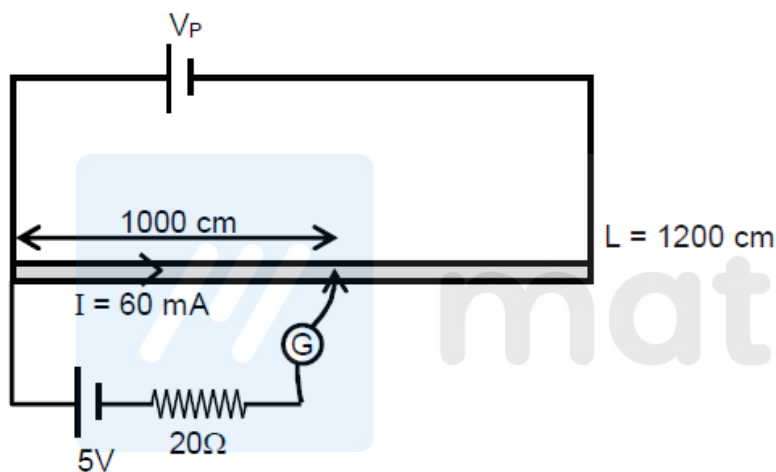
एक विभवमापी तार जिसकी लम्बाई 1200 cm है तथा इसमें धारा 60 mA बह रही है। यदि एक 5V विद्युत वाहक बल की बैटरी जिसका आन्तरीक प्रतिरोध 20Ω है की विभवमापी तार पर संतुलित लम्बाई 1000 cm है। विभवमापी का प्रतिरोध ज्ञात करो।

- (1) 80Ω (2) 100Ω (3) 120Ω (4) 60Ω

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(2)



$$\text{Potential gradient विभव पतन} = \frac{5}{1000} = \frac{V_P}{1200}$$

$$V_P = 6V$$

$$\text{and तथा } R_P = \frac{V_P}{I} = \frac{6}{60 \times 10^{-3}} = 100\Omega$$

Q4

In full scale deflection current in galvanometer of 100Ω resistance is 1 mA. Resistance required in series to convert it into voltmeter of range 10 V.

गैल्वेनोमीटर जिसका प्रतिरोध 100Ω है, में अधिकतम परास के लिए धारा 1 mA है तो इसे 10 V परास के वोल्टमीटर में बदलने के लिए श्रेणीक्रम में आवश्यक प्रतिरोध ज्ञात कीजिए।

- (1) $0.99\text{ K}\Omega$ (2) $9.9\text{ K}\Omega$ (3) $9.8\text{ K}\Omega$ (4) $10\text{ K}\Omega$

8th Jan Evening

Sol

(2)

$$V_g = i_g R_g = 0.1 \text{ V}$$

$$V = 10 \text{ V}$$

$$R = R_g \left(\frac{V}{V_g} - 1 \right)$$

$$= 100 \times 99 = 9.9 \text{ K}\Omega$$

Q5

A screw gauge advances by 3mm in 6 rotations. There are 50 divisions on circular scale. Find least count of screw gauge ?

एक स्कू गेज 6 चक्रण पर 3mm आगे बढ़ता है। वृत्तीय पैमाने पर 50 भाग है। स्कू गेज का अल्पतमांक ज्ञात करो—

(1) 0.002 cm

(2) 0.001 cm

(3) 0.01 cm

(4) 0.02 cm

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Sol

(2)

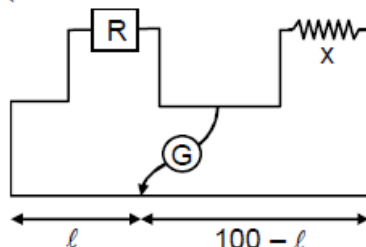
$$\text{Pitch पिच} = \frac{3}{6} = 0.5 \text{ mm}$$

$$\text{L.C.} = \frac{0.5 \text{ mm}}{50} = \frac{1}{100} \text{ mm} = 0.01 \text{ mm} = 0.001 \text{ cm}$$

Q6

If in a meter bridge experiment, the balancing length ℓ was 25 cm for the situation shown in the figure. If the length and diameter of the wire of resistance R is made half, then find the new balancing length in centimetre is

दर्शाये गये मीटर सेतु प्रयोग में सन्तुलन लम्बाई ℓ का मान 25 cm है, यदि प्रतिरोध R की लम्बाई तथा व्यास को आधा कर दिया जाये तो नयी सन्तुलन लम्बाई सेमी में ज्ञात करो—

9th Jan Evening

40.00

$$\frac{X}{R} = \frac{75}{25} = 3$$

$$R = \frac{\rho \ell}{A} = \frac{4\rho \ell}{\pi d^2}$$

$$R' = \frac{4\rho \left(\frac{\ell}{2}\right)}{\pi \left(\frac{d}{2}\right)^2} = 2R$$

$$\text{then } \frac{X}{R'} = \left(\frac{100 - \ell}{\ell}\right)$$

$$\frac{100 - \ell}{\ell} = \frac{X}{2R} = \frac{3}{2}$$

$$\ell = 40.00 \text{ cm}$$



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